

Problem 1

We are asked to test whether the AI loan approval rate differs significantly from the manual officer approval rate.

Given data

Approval data	Sample Size	Approval Rate
AI	$n_1 = 300$	71%
Manual	$n_2 = 280$	66%

Significant level $\alpha = 0.05$

a. Is there a significant difference?

$$(n_1 p_1, n_1 (1 - p_1) > 5, n_2 p_2, n_2 (1 - p_2) > 5)$$

So Z-test for two proportions

Why?

We are comparing two independent proportions

Step 2 :-

- Null hypothesis $H_0: p_{AI} = p_{\text{manual}}$
- Alternative hypothesis $H_1: p_{AI} \neq p_{\text{manual}}$

Step 3 :-

Pooled proportion

$$p_{\text{pooled}} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{(0.71 \times 300) + (0.66 \times 280)}{300 + 280}$$

$$p_{\text{pooled}} = \frac{398}{580}$$

Step 4:- Standard error

$$SE = \sqrt{p_{pooled} (1 - p_{pooled}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}$$

$$= \sqrt{0.68 * (1 - 0.68) \left(\frac{1}{300} + \frac{1}{280} \right)}$$

$$SE = \sqrt{0.001485}$$

Step 5:- Z score

$$Z = \frac{p_1 - p_2}{SE}$$

$$= \frac{0.71 - 0.66}{0.03854}$$

$$= \frac{0.05}{0.038}$$

Step 6:- p-value

$$p\text{-value} = 2 * P(Z > 1.296)$$

$$= \underline{\underline{0.097}}$$

$$p\text{-value} = 2 * 0.097$$

$$= \underline{\underline{0.194}}$$

Step 7:- $\alpha = 0.05$

$$p\text{-value} \approx 0.194 > 0.05$$

b. Does this indicate bias or improvement?

- Data ethics viewpoint:

The AI approval rate is slightly higher 71% v 66%
is not statistically significant

- Before labelling AI as fair

whether approvals differ by protected attributes

loan risks levels → performance evaluation

c. Additionally tests

- Fairness Analysis

Test for bias against protected groups.

- Confusion Matrix & ROC Analysis

Compare AI predictions vs actual loan repayment

- Sensitivity Analysis

check if AI behave consistently under slightly

- A/B Testing in Real Environments

Compare AI decisions to manual officer

- Explainability checks.

Ensure AI decisions are interpretable

Problem 2

a. Form Simple Linear Regression

$$\hat{y} = b_0 + b_1 x$$

$$b_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

$$\text{mean} = \bar{x} = \frac{\sum x_i}{n} \quad \bar{y} = \frac{\sum y_i}{n}$$

$$\text{Sum of } x = 1+2+3+4+5+1+2+3+4+5 = \frac{30}{10} = 3$$

$$\text{Sum of } y = 7.5+6.8+6.0+5.2+4.6+7.8+6.9+6.1+5.4+4.7 = 61.0$$

$$\bar{y} = \frac{61}{10} = \underline{\underline{6.1}}$$

we need $\sum (x_i - \bar{x})(y_i - \bar{y})$ & $\sum (x_i - \bar{x})^2$

Car	x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$	$(x - \bar{x})^2$
1	1	7.5	-2	1.4	-2.8	4
2	2	6.8	-1	0.7	-0.7	1
3	3	6.0	0	-0.1	0	0
4	4	5.2	1	-0.9	-0.9	1
5	5	4.6	2	-1.5	-3.0	4
6	1	7.8	-2	1.7	-3.4	4
7	2	6.9	-1	0.8	-0.8	1
8	3	6.1	0	0	0	0
9	4	5.4	1	-0.7	-0.7	1
10	5	4.7	2	-1.4	-2.8	4

$$\text{Sum of } (x - \bar{x})(y - \bar{y}) = -2.8 - 0.7 + 0 - 0.9 - 3.0 - 3.4 - 0.8 + 0 - 0.7$$

$$= -2.8$$

$$= \underline{\underline{-14.1}}$$

$$\text{Sum of } (x - \bar{x})^2 = 4 + 1 + 0 + 1 + 4 + 4 + 1 + 0 + 1 + 4$$

$$= \underline{\underline{20}}$$

$$b_1 = \frac{-14.1}{20}$$

$$= \underline{\underline{-0.705}}$$

Step 3:- compute b_1

$$b_0 = \bar{y} - b_1 \bar{x}$$

$$= 6.1 - (-0.705 * 3)$$

$$= 6.1 + 2.115$$

$$= \underline{\underline{8.215}}$$

Step 4:- Regression Equation

$$\hat{y} = 8.215 - 0.705x$$

b) Scatter plot and Regression line

X axis Age of cars.

Y axis Resale of price

plot the points (x, y)

Draw the regression line

$$\hat{y} = 8.215 - 0.705x$$

The line goes downward.

c) Interpret the slope

$$\text{Slope } b_1 = -0.705$$

Interpretation: For every additional year of age the resale price decreases by ₹ 0.705 lakhs on average

d) predict resale price for age 3.5 years.

$$\hat{y} = 8.215 - 0.705(3.5)$$

$$= 8.215 - 2.4675$$

$$= \underline{5.7475}$$

$$\Rightarrow \boxed{\text{₹ } 5.72 \text{ lakhs (approximately)}}$$

e. Residual for Car No 4

$$\text{Car 4: } x = 4 \quad y = 5.2$$

$$\text{predicted } \hat{y} = 8.215 - 0.705 \times 4$$

$$= \underline{5.395}$$

$$\text{Residual} = y - \hat{y}$$

$$5.2 - 5.395$$

$$= \boxed{-0.195}$$

f. Models limitations & Improvements

Limitations:

- only considers age, ignores other factors, mileage, brand condition
- linear model assumes constant depreciation

Limitations

- Small sample size ($n=10$) $\rightarrow$ may not generalize well
- Residuals may not be perfectly random

Improvements

1. Add more predictors mileage, fuel, type, brand
2. Use multiple regression or non linear regression
3. Increases sample size for more reliable estimates
4. Check residuals for randomness, apply transformation if needed